

ONTARIO, OR, USA

WMO: 726837

Lat: 44.021N

Lon: 117.013W

Elev: 668

StdP: 93.55

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 24162

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-10.6	-17.5	0.9	-13.1	-14.0	1.2	-8.8	10.9	1.0	9.0	1.5	1.2	250	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	37.1	18.8	35.4	18.2	33.8	17.7	20.4	33.2	19.5	32.3	18.6	31.4	2.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.5	24.7	14.7	11.3	24.0	13.5	10.5	23.5	61.3	33.3	58.0	32.4	55.0	31.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
10.7	9.0	7.8	-16.6	39.9	4.7	1.8	-20.0	41.2	-22.8	42.3	-25.5	43.3	-28.9	44.6		
			-16.7	22.8	4.8	1.5	-20.2	23.8	-23.0	24.7	-25.7	25.5	-29.2	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.3	2.2	7.1	10.6	15.7	20.4	25.6	23.9	18.2	10.7	3.9	-1.1	(6)
(7)		DBStd	9.85	4.72	3.56	3.32	3.60	4.18	4.01	3.10	2.92	3.65	3.71	4.10	4.74	(7)
(8)		HDD10.0	1275	352	219	100	35	5	0	0	0	35	186	343		(8)
(9)		HDD18.3	3092	610	452	348	231	103	24	1	2	47	239	434	601	(9)
(10)		CDD10.0	1777	0	1	11	54	182	312	483	432	246	55	2	0	(10)
(11)		CDD18.3	552	0	0	0	1	21	86	226	176	42	1	0	0	(11)
(12)		CDH23.3	7654	0	0	1	37	374	1172	3037	2303	697	35	0	0	(12)
(13)		CDH26.7	3885	0	0	0	5	118	531	1729	1223	273	7	0	0	(13)
(14)	Wind	WSAvg	2.5	1.8	2.4	2.9	3.4	3.1	2.6	2.3	2.2	2.2	2.0	2.0	(14)	
(15)	Precipitation	PrecAvg	250	34	23	24	19	23	17	6	8	10	15	32	37	(15)
(16)		PrecMax	422	94	59	94	73	139	59	30	54	49	53	100	113	(16)
(17)		PrecMin	124	0	2	0	1	0	0	0	0	0	0	2	2	(17)
(18)		PrecStd	63	17	16	18	13	25	14	8	13	11	13	20	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	14.6	21.3	27.2	32.8	37.3	39.8	37.9	34.2	27.4	17.2	11.4	(19)
(20)		MCWB	5.2	8.3	10.9	14.3	17.0	19.0	19.3	18.6	17.4	14.4	10.0	7.2		(20)
(21)		2%	DB	7.4	12.1	18.5	23.8	29.8	34.0	37.8	36.2	32.1	23.0	14.1	8.9	(21)
(22)		MCWB	4.2	6.7	9.7	12.5	15.4	17.5	19.1	18.3	16.7	12.8	8.6	5.6		(22)
(23)		5%	DB	5.9	9.9	16.3	21.4	27.5	32.1	36.3	34.6	29.9	20.7	12.2	7.0	(23)
(24)		MCWB	3.3	5.6	8.6	11.2	14.7	16.9	18.6	17.7	15.8	11.4	7.4	4.5		(24)
(25)		10%	DB	3.9	7.8	14.0	18.7	25.1	29.8	34.5	32.9	27.7	18.1	10.2	4.9	(25)
(26)		MCWB	2.0	4.3	7.6	10.0	13.7	15.7	18.1	17.4	15.1	10.3	6.4	2.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	9.3	11.9	15.0	18.0	20.7	22.3	20.9	19.6	15.5	11.3	8.1	(27)
(28)		MCDWB	8.2	13.8	17.6	25.3	29.9	34.4	35.9	33.2	29.3	23.5	15.2	10.3		(28)
(29)		2%	WB	4.6	7.2	10.6	13.1	16.5	19.0	20.8	19.8	18.2	13.7	9.6	6.3	(29)
(30)		MCDWB	6.8	11.0	16.5	21.9	27.4	31.4	33.7	32.5	28.5	21.5	13.3	8.5		(30)
(31)		5%	WB	3.4	5.9	9.6	11.8	15.4	17.8	19.9	18.9	17.0	12.3	8.0	4.6	(31)
(32)		MCDWB	5.4	9.3	15.2	19.1	25.6	29.7	33.1	31.8	27.2	19.2	11.2	6.7		(32)
(33)		10%	WB	2.3	4.7	8.3	10.7	14.3	16.7	19.0	18.0	15.7	11.0	6.7	3.0	(33)
(34)		MCDWB	3.9	7.6	13.6	17.9	23.5	27.9	32.2	30.9	26.2	17.0	9.8	4.7		(34)
(35)	Mean Daily Temperature Range	MDBR	6.8	9.2	12.3	13.8	14.6	15.4	17.3	17.6	17.2	14.2	10.4	7.1	(35)	
(36)		5% DB	MCDBR	8.5	12.1	16.1	18.5	18.6	19.1	19.0	19.7	20.2	17.8	12.9	8.3	(36)
(37)		MCWBR	6.0	7.6	8.9	9.2	7.6	7.5	7.4	7.5	8.2	9.0	8.1	5.8	(37)	
(38)		5% WB	MCDBR	7.7	10.6	14.1	16.6	16.7	17.5	18.1	18.5	18.2	15.9	11.6	7.5	(38)
(39)		MCWBR	5.6	6.9	8.2	8.5	7.0	7.3	7.5	7.4	7.7	8.3	7.6	5.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.277	0.292	0.315	0.337	0.335	0.340	0.344	0.321	0.296	0.276	0.269	(40)	
(41)		taud	2.562	2.579	2.551	2.467	2.425	2.456	2.448	2.418	2.483	2.555	2.566	2.553	(41)	
(42)		Ebn at Noon	882	935	956	951	933	932	923	909	908	895	864	846	(42)	
(43)		Edh at Noon	64	75	88	104	112	109	109	109	94	76	63	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.51	2.51	3.80	5.40	6.51	7.43	7.74	6.67	5.12	3.33	1.91	1.24	(44)	
(45)		RadStd	0.24	0.32	0.38	0.42	0.47	0.42	0.28	0.29	0.29	0.27	0.17	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.64	N/A	N/A	+0.93	N/A	N/A	N/A	N/A	+160	+98
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	N/A

Nomenclature: See separate page